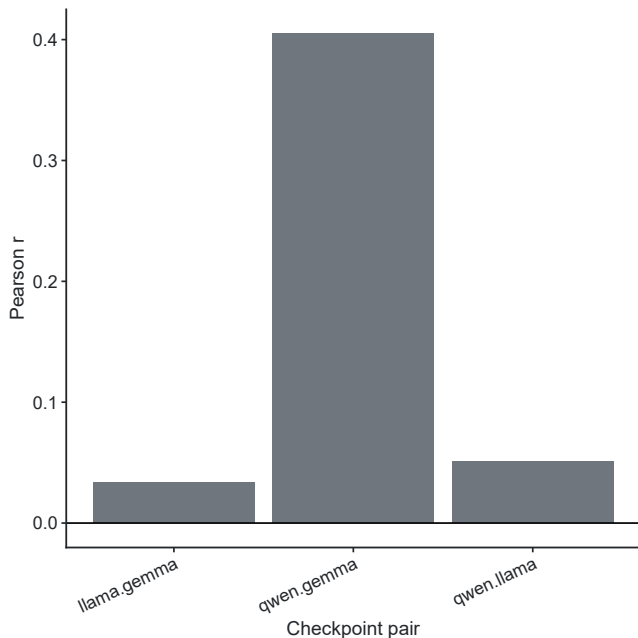


Supplementary Figure S5 | Direction-isomorphism and non-homology controls

Functional recurrence does not imply coordinate or direction identity

a Direction identity is weak across checkpoints



b Functional geometry recurrence depends on objective

